

SpeedCloud®
Supreme[↑]
Performance Benchmark Report



In today's digital-first business landscape, infrastructure performance is no longer just a technical parameter, it directly influences application responsiveness, user experience, scalability, business continuity, analytics processing, and operational efficiency.

To validate the performance capabilities of SpeedCloud Supreme, our premium private cloud offering, a comprehensive benchmark assessment was conducted across the four most critical infrastructure pillars:

Capability Validation Method



These benchmark tests were designed to simulate real-world enterprise workloads and measure the cloud platform's ability to deliver low latency, high throughput, superior responsiveness, and consistent stability under demanding conditions.

1

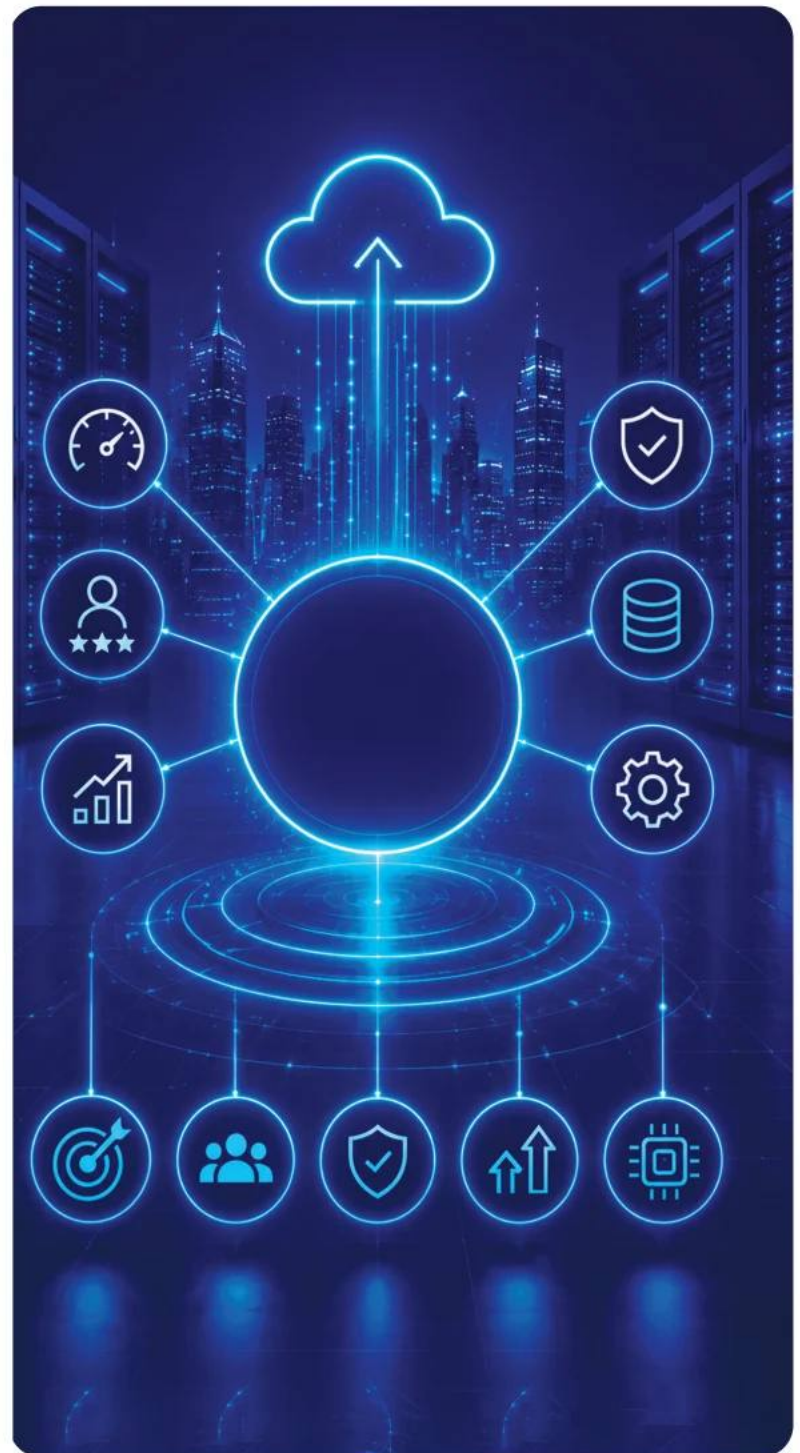
CPU Performance Benchmark – Geekbench 6

Why CPU Benchmarking Matters

The CPU is the processing core of any cloud infrastructure. Whether it is handling enterprise applications, database operations, virtualization workloads, analytics, or multi-user environments, CPU efficiency directly impacts overall performance.

Geekbench 6 evaluates processor performance across:

- Single-core processing capability
- Multi-core scalability
- Real-world workload simulation
- Compression and encryption handling
- AI and compute-intensive operations
- Virtualization efficiency



SpeedCloud[®] Supreme¹ – Geekbench 6 Results

Test	Score
Single-Core Score	2017
Multi-Core Score	3947



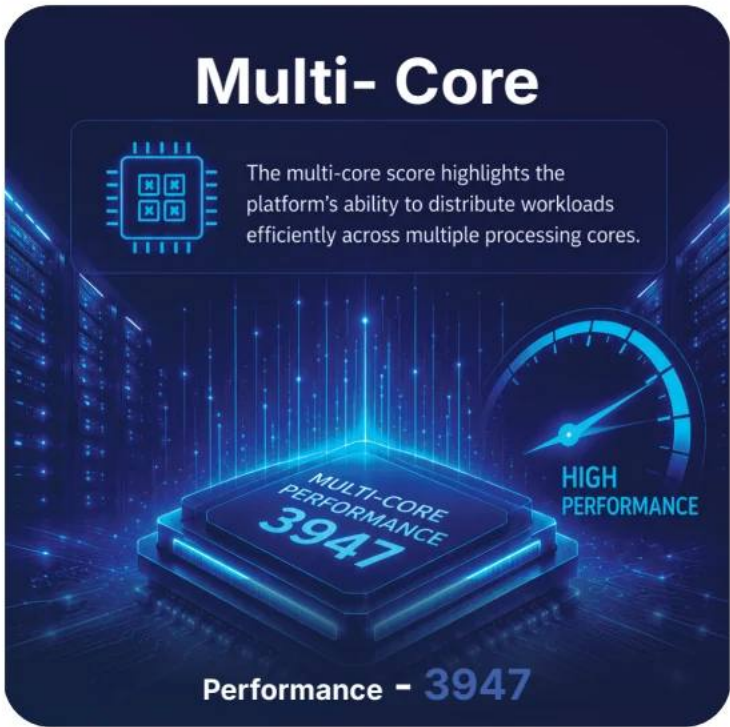
Performance Analysis

Single-Core Performance – 2017

A strong single-core score indicates excellent responsiveness for applications that rely heavily on individual thread performance.

The result demonstrates that SpeedCloud Supreme provides:

- Faster response times
- Better application responsiveness
- Lower execution delay
- Smooth user experience
- Efficient transaction handling



Performance Analysis

Multi-Core Performance – 3947

The multi-core score highlights the platform's ability to distribute workloads efficiently across multiple processing cores.

The benchmark confirms that SpeedCloud Supreme can effectively handle:

- Simultaneous workloads
- Resource-intensive applications
- Large virtual machine deployments
- Enterprise-grade compute requirements
- Efficient transaction handling



Storage Performance Benchmark

FIO

Why Storage Benchmarking Matters

Storage performance is one of the most critical elements of any cloud environment.

The FIO benchmark measures:

- Input/Output Operations Per Second (IOPS)
- Latency
- Throughput
- Random read/write efficiency
- Sequential read/write efficiency

These metrics are essential for evaluating:

- Database performance
- Virtual machine responsiveness
- Application loading speed
- File processing efficiency
- Backup & recovery operations
- Large data workloads

Storage Latency Results

Benchmark Results

Test Type	Avg Latency (µs)	p99 Latency (µs)
Random Read	316	545
Random Read/Write	Read: 349, Write: 934	Read: 603, Write: 2999

Analysis

Random Read Latency – 316 µs Average

The exceptionally low random read latency demonstrates:

- High-speed storage responsiveness
- Fast application access
- Quick database retrieval operations
- Reduced system wait times

The p99 latency remaining within 545 µs indicates highly consistent performance even under load conditions.

Random Read/Write Performance

Mixed workload testing simulates real enterprise scenarios where systems simultaneously perform reads and writes.

The average read latency of **349 µs** indicates that the storage subsystem can retrieve data extremely quickly even during simultaneous write activity. This is critical for applications that require fast access to active datasets, such as databases, ERP platforms and much more.

The average write latency of **934 µs** remains well within enterprise-grade performance levels for mixed workloads, showing the storage system's ability to efficiently handle continuous data writes without creating major bottlenecks.

The p99 latency reflects the worst-case response time experienced by only 1% of operations and is one of the most important indicators of storage consistency under stress.

- **603 µs p99 read latency** demonstrates highly stable read performance even during peak workload conditions.
- **2999 µs p99 write latency** indicates that even under heavy mixed operations, write response times remain controlled and predictable.

The results indicate:

- Stable multi-operation handling
- Efficient storage controller performance
- Enterprise-grade workload balancing
- Strong transactional capability

Storage IOPS Benchmark (4K Block Size)

Benchmark Results

Test Type	IOPS	Bandwidth (MiB/s)	Avg Latency (ms)	p99 Latency
Random Read	94.1k	368	10.87	20
Random Write	61.3k	239	16.7	40
Random Read/Write	Read: 38.6k Write:38.6k	Read: 151 Write:151	Read: 12.78 Write: 3.74	Read: 32 Write:34
Sequential Read	249k	973	4.11	9.37
Sequential Write	74.3k	290	13.77	33

Performance Analysis

Random Read Performance

- 1 Achieving **94.1k random read IOPS** means the storage subsystem can process approximately **94,100 small read operations every second**. This reflects the platform's ability to retrieve scattered data blocks rapidly from storage.
- 2 The accompanying **368 MiB/s bandwidth** indicates that the storage is not only processing a high number of operations but is also transferring a substantial amount of data continuously without bottlenecks.
- 3 The **10.87 ms average latency** shows that response times remain consistently controlled even while handling tens of thousands of read requests simultaneously.
- 4 More importantly, the **20 ms p99 latency** indicates that even during peak load conditions, almost all read requests are completed within a predictable and stable response window. This is critical because enterprise applications are highly sensitive not only to average performance, but also to consistency during workload spikes.

Random Write Performance

- 1 The storage achieved **61.3k random write IOPS**, meaning it can efficiently process over **61,000 small write operations every second**.
- 2 Write operations are generally more demanding than reads because the storage system must commit data safely and maintain consistency. Sustaining this level of write performance demonstrates strong backend storage optimization & controller efficiency.
- 3 The **239 MiB/s bandwidth** confirms the platform's ability to continuously absorb large volumes of incoming data without congestion.
- 4 The **16.7 ms average latency** indicates stable write response times under sustained activity, while the **40 ms p99 latency** shows that even peak write operations remain controlled and predictable under stress conditions.

Random Read/Write Mixed Workload Performance

The mixed workload benchmark simulates real enterprise environments where systems continuously perform both read and write operations together

1 The Platform Delivered:

38.6k	38.6k
Read IOPS	Write IOPS

This balanced distribution indicates that the storage architecture can maintain stable performance across simultaneous operations instead of favoring only reads or only writes.

2 The 151 MiB/s read and write bandwidth further demonstrates balanced throughput handling across mixed workloads.

3 Latency Also Remained Highly Controlled:

12.78 ms	13.74 ms
Average Read Latency	Average Read Latency

The closeness of these values indicates excellent workload balancing and minimal contention between read and write operations.

4 The p99 latencies:

32 ms	34 ms
For Read	For Writes

show that even during heavy concurrent workloads, response times remain predictable and enterprise-grade.

Sequential Read Performance

1	2	3
The sequential read benchmark achieved an exceptionally high 249k IOPS with 973 MiB/s throughput. Sequential operations occur when data is accessed continuously in order, which is common in: • Backups • Media streaming • Large file transfers • Analytics processing	The near 1 GiB/s bandwidth indicates that SpeedCloud Supreme is capable of moving massive volumes of data extremely quickly.	The 4.11 ms average latency and 9.37 ms p99 latency are particularly impressive because they show that the platform maintains extremely fast response times even while processing very large continuous data streams.

Sequential Write Performance

1

The sequential write benchmark achieved:

74.3k

IOPS

290 MiB/s

Throughput

This demonstrates strong capability for sustained continuous data writing.

2

The **13.77 ms average latency** indicates stable write handling, while the **33 ms p99 latency** confirms predictable performance even during extended write-intensive operations.

Storage Throughput Results (64K Block Size)

The 64K block size benchmark measures large-block data transfer performance, which is critical for workloads involving continuous data movement, bulk transfers, backups, analytics, media processing, and high-throughput enterprise applications.

Unlike 4K benchmarks that focus on transactional responsiveness, the 64K benchmark evaluates how efficiently the storage subsystem handles larger volumes of data over sustained operations.

Test Type	Bandwidth (MiB/s)	IOPS	Avg Latency (ms)
Random Read	5919	23.7k	10.79
Random Write	1748	6990	36.61
Random Read/Write	Read: 1231 Write: 1241	Read: 4924 Write: 4964	Read: 26.95 Write: 24.81
Sequential Read	6439	25.8k	9.92
Sequential Write	2499	9994	25.61

Random Read Throughput Performance

1 The random read benchmark achieved:

5919 MiB/s bandwidth	23.7k IOPS	10.79 ms Average Latency
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This indicates that the storage subsystem can retrieve large volumes of randomly distributed data at extremely high speeds.

2 The exceptionally high **5919 MiB/s throughput** demonstrates that SpeedCloud Supreme is optimized for fast data access across large datasets without creating major storage bottlenecks.

3 Although the IOPS count is lower compared to the 4K benchmark, this is expected because each operation processes a much larger 64K data block. In practical terms, fewer operations are needed to transfer significantly larger amounts of data.

4 The controlled latency of **10.79 ms** confirms that throughput remains stable without introducing excessive response delays.

Random Write Throughput Performance

1 The random write benchmark delivered:

1748 MiB/s bandwidth	6990 IOPS	36.61 ms Average Latency
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Write throughput is naturally more demanding because data must be committed reliably to storage while maintaining consistency and integrity.

2 Achieving nearly **1.7 GiB/s write throughput** demonstrates strong storage controller efficiency and backend optimization for sustained write-heavy workloads.

3 The higher latency compared to reads reflects the additional processing involved in safely writing data, but the response times remain within enterprise-grade operational limits for high-throughput write operations.

Random Read/Write Mixed Throughput Performance

1

The Platform Achieved:

1231 MiB/s	1241 MiB/s	4924	4964
Read bandwidth	Write bandwidth	Read IOPS	Write IOPS

The near-identical read and write throughput values demonstrate excellent workload balancing and efficient simultaneous operation handling.

2

Latency Values Remained Stable:

26.95 ms	24.81 ms
Read Latency	Write Latency

This indicates that neither operation type heavily impacts the other, which is a strong sign of a well-optimized enterprise storage architecture.

Sequential Read Throughput Performance

1

The sequential read benchmark achieved:

6439 MiB/s	25.8k	9.92ms
bandwidth	IOPS	Average Latency

This is one of the strongest indicators of high-throughput storage capability.

2

The exceptionally high 6.4 GiB/s throughput confirms that SpeedCloud Supreme is engineered for enterprise-scale data movement with minimal bottlenecks.

3

The low latency further indicates that throughput performance is achieved without sacrificing responsiveness.



Network Performance Benchmark iPerf

Why Network Benchmarking Matters

In modern cloud infrastructure, network performance is as important as compute and storage.

A high-performance network ensures:

- Faster application access
- Stable remote connectivity
- Efficient data transfer
- Smooth cloud communication
- Low-latency operations
- Better user experience

The iPerf benchmark evaluates:

- Network throughput
- Bandwidth stability
- Transfer efficiency
- Data consistency

SpeedCloud Supreme – iPerf Results

Test	Transfer (Sender in GB)	Bitrate (Sender in Gbits/sec)	Transfer (Sender in GB)	Bitrate (Receiver in Gbits/sec)
Test 1	35.1	30.2	35.1	30.2
Test 1	35.7	30.7	35.7	30.7
Test 1	36.1	31	36.1	31

Test 1 – 30.2 Gbits/sec Throughput

During the first test, the network achieved:

35.1 GB

Data Transfer

30.2 Gbits/sec

Sustained Throughput

This demonstrates that SpeedCloud Supreme can consistently move very large volumes of data at extremely high speeds without significant network degradation.

The identical sender and receiver throughput values also indicate:

- | | |
|-------------------------------|-------------------------------------|
| ↗ Minimal packet loss | ↗ Efficient network synchronization |
| ↗ Stable transmission quality | ↗ Strong network reliability |

Test 2 – 30.7 Gbits/sec Throughput

The second test achieved:

35.7 GB	30.7 Gbits/sec
Data Transfer	Throughput

The increase in throughput demonstrates stable scaling capability under continuous network load.

Maintaining consistent sender and receiver rates confirms:

- | | |
|---|-------------------------------------|
| ↗ Network stability under sustained traffic | ↗ Efficient bandwidth utilization |
| ↗ Low transmission overhead | ↗ Reliable high-speed communication |

Test 3 – 31 Gbits/sec Throughput

The third benchmark achieved:

36.1 GB	31 Gbits/sec
Transfer	Throughput

Crossing the 31 Gbps mark highlights the capability of SpeedCloud Supreme to sustain enterprise-grade high-bandwidth workloads consistently.

The matching sender and receiver performance further validates:

- | | |
|---|----------------------------------|
| ↗ Excellent packet delivery consistency | ↗ Optimized network architecture |
| ↗ Stable end-to-end communication | ↗ Efficient throughput handling |

The iPerf benchmark results collectively demonstrate that SpeedCloud Supreme delivers:

- | | |
|---|--|
| ↗ Extremely high network throughput | ↗ Low network bottlenecks |
| ↗ Stable bandwidth consistency | ↗ Enterprise-grade communication performance |
| ↗ Reliable large-scale data transfer capability | |

The ability to consistently sustain approximately 30–31 Gbits/sec throughput across multiple test runs confirms that the network architecture is designed for high-performance enterprise workloads.



Memory Performance Benchmark Sysbench

Why Memory Benchmarking Matters

Memory performance plays a critical role in:

- Application responsiveness
- Data caching
- VM efficiency
- Database operations
- Analytics workloads
- Real-time processing

SpeedCloud Supreme - Sysbench Results

Metric	Result
Data Transferred	76597.22 MiB
Transfer Rate	7658.72 MiB/sec
Maximum Latency	0.45 ms

1 Large Memory Data Handling - 76597.22 MiB Transferred

During the benchmark, the memory subsystem successfully processed over 76 GB of data continuously, which demonstrates the platform's ability to sustain large-scale memory operations without instability or performance degradation.

2 High Memory Throughput – 7658.72 MiB/sec

Achieving a transfer rate of over 7.6 GiB/sec indicates that SpeedCloud Supreme can move very large amounts of data through memory extremely quickly.

This is particularly important because memory acts as the active workspace for applications and virtual machines. Faster memory throughput allows workloads to process data more efficiently without waiting for memory operations to complete.

3 Ultra-Low Memory Latency – 0.45 ms

The maximum latency recorded during the benchmark was only 0.45 milliseconds, which indicates extremely fast memory response times.

Low memory latency is critical because modern enterprise applications constantly access memory for active computations and transaction processing. Delays at the memory layer can directly impact application speed and user experience.

The Sysbench benchmark results collectively demonstrate that SpeedCloud Supreme delivers:

- High-speed memory processing
- Extremely low-latency responsiveness
- Efficient large-scale workload handling
- Strong virtualization support
- Stable multi-user performance



Overall Benchmark Summary

Performance Highlights

Compute Excellence

- Strong single-core and multi-core CPU performance
- Optimized for enterprise applications and virtualization
- Efficient multi-user workload handling

High-Speed Networking

- ~31 Gbps network throughput
- Stable transfer consistency
- Enterprise-grade bandwidth capability

The benchmark results validate that SpeedCloud Supreme is engineered to deliver premium-grade private cloud performance capable of supporting demanding enterprise workloads with consistency and efficiency.

These results reflect a cloud platform designed not only for performance, but also for:

Enterprise Storage Architecture

- High IOPS capability
- Low-latency storage performance
- Fast sequential throughput
- Stable random workload execution

High-Performance Memory

- 7.6+ GB/sec memory transfer rate
- Ultra-low latency response
- Optimized workload execution

- Business continuity
- Scalability
- Application stability
- Operational efficiency
- Long-term reliability

About NxtGen

NxtGen is India's largest sovereign cloud and AI provider, serving more than 1000+ customers across the country. NxtGen operates SpeedCloud[®]- public cloud platform, SpeedCloud Supreme-an alternative to private cloud and Industry vertical clouds for the Government and Financial Services. NxtGen delivers value to its customers by ensuring superior performance at lower cost. NxtGen AI portfolio unifies GPU platforms from NVIDIA, AMD & Intel and provides a wide choice of highly optimized models to build enterprise use cases. NxtGen is working to truly democratize the cloud and AI technologies, by making them viable for enterprises of all sizes.